

Hall Effect Guide

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In this experiment we will measure Hall voltage and resistivity of a semiconductor. From these measurements we will calculate the density of charge carriers in the semiconductor, their type, and mobility.

Theory

We first discuss the Hall Effect in the framework of the [Drude model](#). The Hall Effect, discovered by Edwin Hall in 1879, is the appearance of an electric field (the Hall field) in a current carrying conductor placed in an external magnetic field. This Hall field is perpendicular both to the current direction and the magnetic field direction.

Conductivity of a Semiconductor

The conductivity of a semiconductor is due to the movement of negatively charged electrons and/or positively “charged” holes (a hole is a “missing” electron in the semiconductor). In thermal equilibrium their densities obey the so-called mass action law:

$$n \cdot p = n_i^2 \quad (1)$$

where n is the electrons’ concentration, p is the holes’ concentration, and n_i is the temperature-dependent intrinsic carrier density of the semiconductor. In Germanium, the semiconductor you are going to measure, $n_i \approx 2.5 \times 10^{13} \text{ cm}^{-3}$ at room temperature. In an intrinsic (without dopants) semiconductor $n = p = n_i$. In a doped semiconductor at moderate temperatures above the dopant binding energy, but well below the forbidden gap energy, the density of the majority carriers is equal to the doping and the density of the minority ones can be derived from (1).

In an intrinsic semiconductor the charge carriers appear due to excitation across the gap. This causes the density to vary with the temperature roughly according to

$$n_i \propto \exp\left(-\frac{E_g}{2k_B T}\right), \quad (2)$$

where E_g is the energy gap between the conduction and the valence bands, k_B is the Boltzmann constant and T denotes the temperature.

Example: The intrinsic density in silicon at room temperature is: $n_i = 1.4 \times 10^{10} \text{ cm}^{-3}$. With donor doping of $N_d = 7 \times 10^{16} \text{ cm}^{-3}$, the electron and hole densities are $n \approx N_d = 7 \times 10^{16} \text{ cm}^{-3}$ and $p = \frac{n_i^2}{n} = 2800 \text{ cm}^{-3}$.

The semiconductor's conductivity depends on the temperature. This dependency can be roughly separated into three regions: Freeze out (Ionization), Extrinsic and Intrinsic (Fig.1).

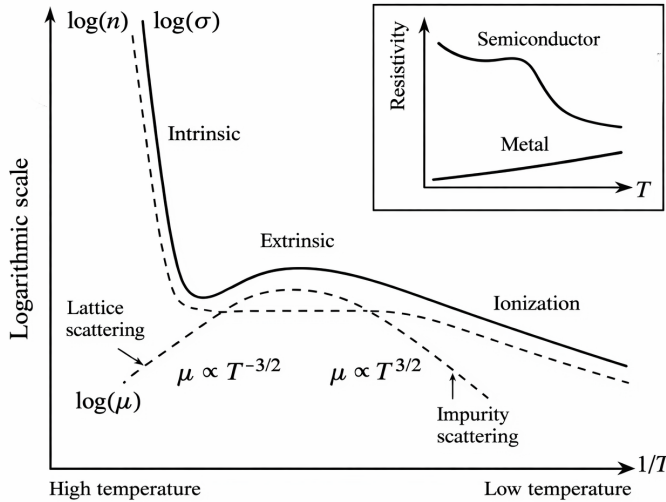


Figure 1: Schematic temperature dependence of conductivity, charge-carriers mobility and density in a n-type semiconductor. Note that the temperature dependence of the mobility is just an illustration, and is different for electrons and holes, and for different semiconductors. Image gratis: [Kasap(2006), Ch.5, p.405]

At **freeze-out** temperatures the carrier density drops with cooling, and eventually the semiconductor becomes an insulator.

As the temperature rises the dopants become ionized. Once they are all ionized the density of the majority carriers remains almost independent on temperature, while the density of the minority ones increases. This temperature range is called the **extrinsic** region.

As the temperature increases even more, as $n_i \propto \exp\left(-\frac{E_g}{2k_B T}\right)$ also increases and eventually becomes comparable or even larger than the doping concentration. Such temperature region is called the **intrinsic**. Note that there are no well-defined temperatures which separate the regions, rather this separation reflects approximations which are typically made to find n and p without solving nonlinear equations.

Temperature dependence of the conductivity at freezeout and the intrinsic regions is determined mostly by change in the carrier density. In contrast, in the extrinsic region the majority carrier density is approximately constant, and the change of the carriers mobility determines the conductivity temperature dependence. At low temperatures, the carriers are scattered mostly by ionized dopants, while at high temperatures they are mostly scattered by phonons in the crystal. In the Drude model, the scattering is accounted for through relaxation time τ ; the carriers' mobility is proportional to it.

The Hall Effect

Consider a conductor in Fig. 2 in magnetic field $\vec{B} = B\hat{z}$. Let the conductor carry a current $\vec{I} = I\hat{x}$ in the direction $\hat{x}$. The Lorentz force acting on a moving charge carrier is

$$\vec{F}_B = q(\vec{v} \times \vec{B}) = -qvB\hat{y}, \quad (3)$$

where q is its electric charge and v is the velocity of the charged carrier. We used the fact that v is in $\hat{x}$ direction. **Both v and q can be positive or negative.**

Let us consider positively charged carriers as an example. In this case $\vec{F}_B$, pushes the charges down. Charge accumulation, in turn, induces a Hall electric field $\vec{E}_H = E_H\hat{y}$. This electric field applies a force on the charge carriers in direction $\hat{y}$, so that in equilibrium there is no net current in $\hat{y}$ direction. The condition for this is (4)

$$\vec{F}_B = -q\vec{E}_H \quad (4)$$

This field creates a potential difference between the points with the same x on the lower and upper edges of the sample (we assumed here and below that the sample is long and uniform, so the current flows in $\hat{x}$ direction everywhere through the relevant cross section). This potential difference is called the Hall voltage U_H . This treatment is simplistic, and the velocity that appears in the Drude model is the average or the so-called *drift velocity*. But, since the Eq. (4) should hold on average, it gives the correct classical answer.

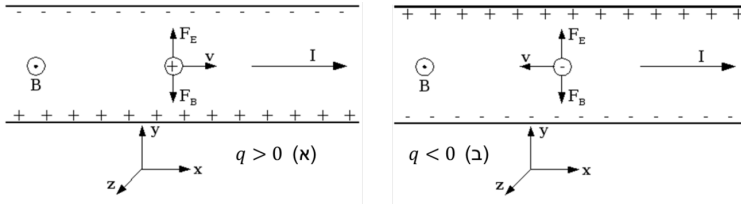


Figure 2: Illustration of the Hall Effect for positive and negative charges. In the steady state the Lorentz force and the Hall electric field compensate each other and no additional charge is accumulated.

For a long rectangular conductor carrying current due to only one type of charge carriers, one can get from Eqs. 3 and 4 the magnitude of the Hall voltage:

$$U_H = \frac{IB}{nqd} \quad (5)$$

where n is the density of the charge carriers and d is the thickness of the conductor. The Hall voltage may be positive or negative depending on the sign of the carrier's charge.

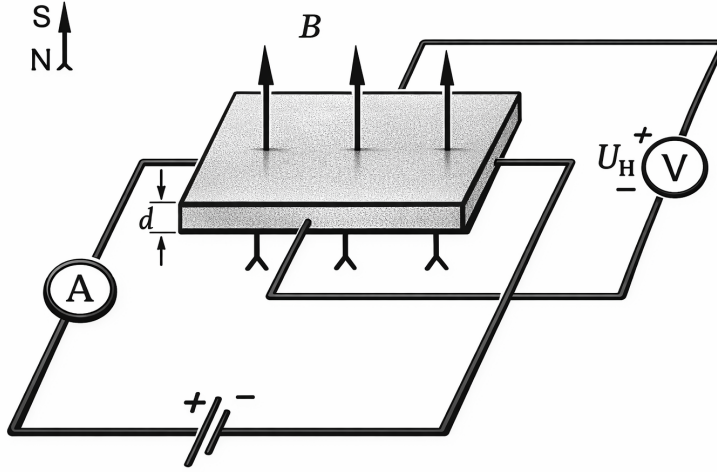


Figure 3: The Hall Effect in a rectangular sample.

We define a parameter called the Hall coefficient R_H as:

$$R_H = \frac{U_H d}{IB} = \frac{1}{nq} \quad (6)$$

Thus, Hall coefficient in the simplest case is set by the charge density and sign.

Generalization of the Hall Effect for more than one carrier type

At relatively high temperatures a semiconductor is in the intrinsic region, and the current is carried by both electrons and holes. This regime is relevant to the experiment you are going to perform; the Hall effect in this regime is more involved, and we discuss it below in some detail.

We shall start from expressing Hall effect in tensor form. For this we introduce resistivity tensor. We shall assume both current density $\vec{j}$ and electric field $\vec{E}$ to be in $x - y$ plane, and therefore we can express the relation between them as:

$$\vec{E} = \hat{\rho} \vec{j} \quad (7)$$

with $\hat{\rho}$ being resistivity tensor:

$$\hat{\rho} = \begin{pmatrix} \rho_{xx} & \rho_{xy} \\ \rho_{yx} & \rho_{yy} \end{pmatrix}. \quad (8)$$

For materials with sufficiently high symmetry (and Ge you will be measuring does possess such symmetry) one can show that $\rho_{xx} = \rho_{yy}$ and $\rho_{yx} = -\rho_{xy}$.

In the simplest case of an isotropic material in zero magnetic field $\hat{\rho}$ is diagonal:

$$\hat{\rho} = \rho_0 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \rho_0 = R_0 \cdot d \cdot W/L, \quad (9)$$

where W is the width, L the length, and d the thickness of the sample, and $R_0 = U_c/I$ is zero magnetic field two terminal resistance of the sample, determined from the voltage across the current contacts of the sample.

Eqs. (7) and (8) are local expression for ohmic conductance and the Hall effect, with $\rho_{xy} = -\rho_{yx}$ being related to the Hall coefficient:

$$\rho_{xy} = R_H B = \frac{B}{nq}. \quad (10)$$

All the carriers are subjected to the same electric field, and the resulting currents are summed up. To facilitate the treatment, we need to invert Eq. (7):

$$\vec{J} = \hat{\sigma} \vec{E}, \quad \hat{\sigma} = \hat{\rho}^{-1}. \quad (11)$$

We defined the conductivity tensor $\hat{\sigma}$. Pay attention to matrix inversion in (11).

In the [Drude theory](#)

$$\hat{\sigma} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{pmatrix} = \frac{\sigma_0}{1 + (\mu B)^2} \begin{pmatrix} 1 & -\frac{\mu B q}{|q|} \\ \frac{\mu B q}{|q|} & 1 \end{pmatrix}, \quad (12)$$

where $\sigma_0 = n\mu|q|$. We introduced mobility μ , which is the proportionality between the electric field acting on carriers and their drift velocity. Its units are $m^2/V \cdot s$.

$$v = \mu E = \frac{\sigma}{n|q|} E \rightarrow \mu = \frac{\sigma}{n|q|} \quad (13)$$

where σ (in units of $1/\Omega m$) is the conductivity.

To treat the case of *both electrons and holes* carrying current together, we shall introduce current densities $\vec{J}_{e,h}$ conductivity tensor $\hat{\sigma}_{e,h}$, resistivity $\hat{\rho}_{e,h}$, and carrier mobility $\mu_{e,h}$ for electrons and holes respectively.

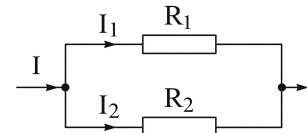
The total current through the sample is:

$$\vec{J} = \vec{J}_e + \vec{J}_h = \hat{\sigma}_e \vec{E} + \hat{\sigma}_h \vec{E} \rightarrow \hat{\sigma} = \hat{\sigma}_e + \hat{\sigma}_h \quad (14)$$

An important remark is in place here: due to particular geometry of the commercial samples, we are forced to determine two-terminal resistance from the voltage U_c across the current leads of the sample. It includes contact resistance that can be not small. One would prefer a more accurate 4-terminal measurement scheme, which requires both current leads and potential prods. As the result, we are bound to overestimate ρ_0 , and thus underestimate carriers' mobility μ defined below.

Mobility is usually defined as a positive parameter, and drift direction is determined by the sign of the charge carriers.

We can think of these currents of these two carrier types as flowing in parallel, like a current flowing through parallel resistors.



Since in the experiment we fix a current and measure voltages, we must express the voltages through the resistivity $\hat{\rho} = \hat{\sigma}^{-1}$. One can easily reverse the tensor $\hat{\sigma}$ for the case of just two types of the charge carriers. The resulting expression, however, is rather bulky. Since the experiment we are performing involves rather small magnetic fields ($\mu B \ll 1$), we restrict ourselves to the lowest non-vanishing orders in B , which happen to be the first for the off-diagonal (Hall), and the second for the diagonal components (a question: explain qualitatively why it is the case) The off-diagonal component of ρ gives the Hall coefficient, which in two-carrier types case *should not be interpreted* as in Eq.(6). Instead:

$$\rho_{yx} = -\rho_{xy} = B \cdot \frac{\mu_h^2 p - \mu_e^2 n}{e \cdot (\mu_e n + \mu_h p)^2} = B \cdot R_H. \quad (15)$$

For a p-type sample at elevated temperatures, the Hall coefficient might even turn to zero, which does not mean that the density is infinite.

Preparatory questions

1. Describe:
 - (a) U_p as a function I_x ,
 - (b) U_H as a function I_x ,
 - (c) U_H as a function B ,
 - (d) U_p as a function B .
2. For a semiconductor of thickness $d = 1 \text{ mm}$, placed in a magnetic field $B_z = 300 \text{ mT}$, and carrying a current of $I_x = 30 \text{ mA}$ the Hall voltage of $U_H = -50 \text{ mV}$ between the upper and lower edges of the sample (see Fig. 3) was measured. What is the type and the density of the dominant carriers (in units of cm^{-3})?
3. For an intrinsic semiconductor resistance 37Ω was measured at the room temperature (20°Celsius). What would be the resistance at 140°Celsius , assuming the carrier mobility remains constant?

Performing the Experiment

Getting to Know the Workplace

The workplace consists of PHYWE Hall-effect module with interchangeable semiconductor sample, set of core and coils, teslameter with Hall probe for magnetic field measurements, AC/DC power supply, multimeter and computer application for taking the readings.

The **PHYWE Hall-effect module** (Fig.4) allows you to apply current and heating to the sample and to measure voltage and temperature. The module is powered by 12 V AC; power-in banana sockets are on the back side of the unit. On the front side are banana sockets for voltage readings.

Warning

Do not apply external power to the PHYWE Hall-effect module front-panel voltage measurement sockets. Doing so will immediately damage both the Hall-effect module and the semiconductor sample.

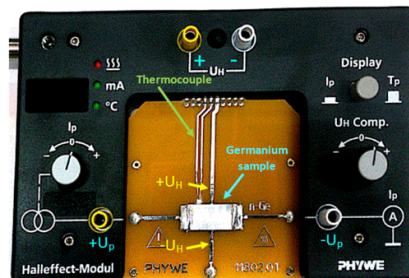


Figure 4: The front side of the module. The yellow connector is connected to the top Hall contact, and the white – to the bottom one. The AC power supply connectors (12 V) and a button to start and stop sample heating are on the back side of the module.



The I_p control knob on the left side of the module sets the current applied to the sample in the range ± 60 mA. The current value is displayed on the unit. The display can be switched to temperature display mode with **Display** button. The temperature of the sample is measured by the thermocouple. The U_H Comp adjustment knob compensates the Hall voltage offset in range of up to ± 10 mV. The U_H and U_p banana sockets are used to measure the corresponding voltage readings with a multimeter. The **On/Off** button on the back side starts heating of the sample. The heating stops automatically at 140°C .

The **set of core, polar pieces and coils** (Fig.5) is used to apply the constant magnetic field B to the sample placed between the po-

lar pieces when powered with DC current. The value of the field B can be varied with **V** and **A** knobs of the power supply.

The magnetometer called **Teslameter** measures the field B value using the **Hall probe**. The probe measures the value and the sign of the magnetic field perpendicular to the probe surface. As written in the manufacturer's **Operation manual**: a positive reading indicates that the field enters the probe from the direction of the surface of the handle that carries the nameplate, whereas a negative value indicates that the field has the opposite direction. When the probe is inserted face side to the front and the coils are wired as shown on the Fig.6, the field is directed from back to front, entering the back side of the probe, and the sign of the field is positive.

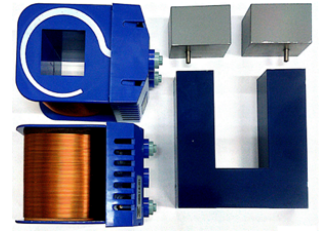


Figure 5: Coils, an iron core and two polar pieces.

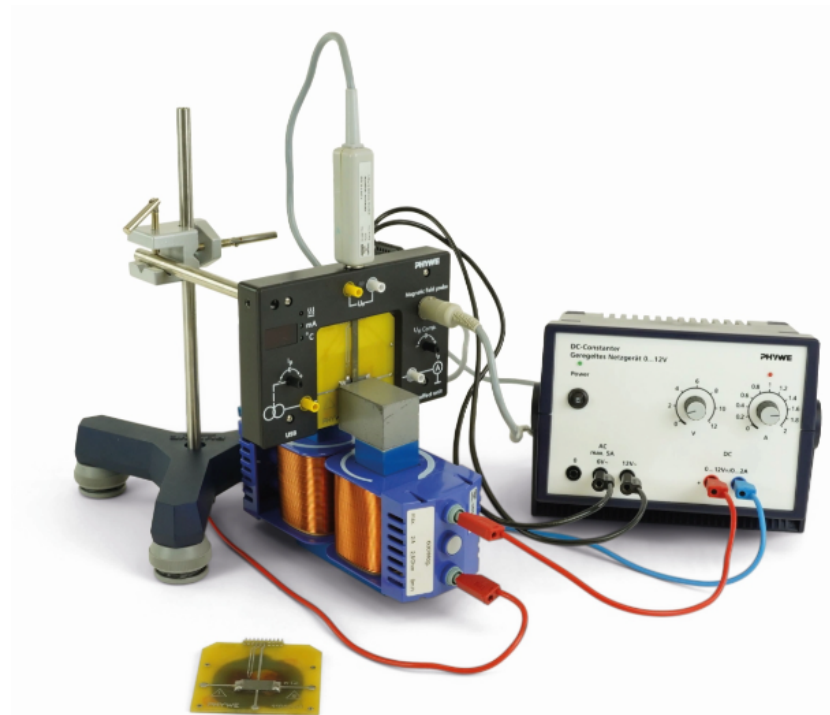


Figure 6: Module with the sample mounted on the coils. Image: PHYWE.

The PHYWE Hall-effect module was modified so that voltages and temperature readings are transmitted to a **computer application**. The interface shows temperature $T(^{\circ}\text{C})$, probe voltage U_p and Hall voltage U_H (mV) and includes a notepad for recording data in a free form. Pressing **SHIFT+ENTER** immediately inserts data readings into the notepad. This is especially convenient when readings change quickly at high temperatures. Notes can be saved or copied for later analysis.

Quantities measured during the experiment

In the experiment we measure the voltages as a function of either B or I . In order to convert them into ρ_{xx} and ρ_{xy} we need to take into account the sample dimensions:

$L = 16.00 \text{ mm}$,
 $W = 10.00 \text{ mm}$.
 $d = 1.00 \text{ mm}$.

$$\begin{aligned} U_p &= \rho_{xx} \frac{L}{Wd} I, \\ U_H &= \rho_{yx} \frac{I}{d} \end{aligned} \quad (16)$$

Here $L = 16.00 \text{ mm}$, $W = 10.00 \text{ mm}$ and $d = 1.00 \text{ mm}$ are the length, width and thickness of the sample.

Misalignment voltage correction

One can note that some samples show non-zero voltage on the Hall probes when $B = 0$, which cannot be explained by our simple model (Eq. 5). This phenomenon is called a misalignment voltage U_m . It happens because of small geometrical misalignment of the top and the bottom Hall contacts in horizontal projection. This voltage is linearly proportional to the probe voltage U_p :

$$U_H = \frac{IB}{nqd}$$

$$U_m = \beta U_p + \text{offset}. \quad (17)$$

Here β is dimensionless small value, easily found in zero-field experiment. We'll perform it further. The offset can be set to zero by adjusting `$U_H$  Comp` knob.

With the misalignment parameter β determined, the corrected Hall voltage U_H can be obtained from the observed voltage on the Hall probes U'_H using a simple linear correction

$$U_H = U'_H - U_m = U'_H - \beta U_p. \quad (18)$$

The corrected Hall voltage now behaves as expected from the Eq. 5.

The Experiment Overview

The experiment consists of the following parts:

#	Part	Result
1	Zero field experiment Measure voltages at $I_p = -30 \text{ mA} \dots 30 \text{ mA}$ at $B = 0$.	Get sample misalignment slope β and sample resistance R .
2	Room temperature experiment At room temperature measure voltages at $I_p = -30 \text{ mA} \dots 30 \text{ mA}$ at $B = 300 \text{ mT}$ and for $B = -300 \text{ mT} \dots 300 \text{ mT}$ at $I_p = 30 \text{ mA}$.	See linear dependence $U_H(I_p \cdot B)$. Sign of slope shows type of carrier: negative for n-type and positive for p-type. Calculate Hall constant R_H , density of charge carriers and Hall mobility μ .
3	Heating experiment Heat the sample to 140°C . Measure voltage-temperature dependence at $B = 300 \text{ mT}$ and $I = 30 \text{ mA}$.	See Hall voltage decreases with temperature for n-Germanium. See Hall voltage crosses zero for p-Germanium. Calculate bandgap energy E_g in electron-volts (eV).

Part 1: Zero field experiment

1. Connect the power supply 12 V AC output to the back of the module.
2. Run the computer application for taking Hall effect measurements. Make sure you understand how to catch readings in the notepad by pressing `SHIFT+ENTER`, how to enter data and comments manually, how to clear the text, copy it and save the text to a file. Write down the sample number and type as a comment, preceded by # sign.
3. Adjust `Ip` knob to get the probe voltage U_p as small as possible (less than 10 mV). Then adjust `UH Comp` knob to get the Hall voltage U_H as small as possible (less than 1 mV). This sets the misalignment offset in Eq.(17) to zero.
4. Record readings for different sample currents I_p ranging from -30 mA to 30 mA in steps of about 5 mA. See a sample in the Table 1.
5. Plot U_H as a function of U_p . If the sample is misaligned, it will show up as a linear trend. Perform the linear regression to find the misalignment slope β .
6. Plot I_p as a function of U_p . Perform the linear regression to find the slope. The slope is equal to conductance $\frac{1}{R}$, giving the resistance R . Note that fitting linear regression of I_p as a function of U_p (not vice versa) satisfies the assumptions of the linear regression model: x -values of the linear regression have negligibly small error compared to y -values (see [Taylor(1996), Ch.8.2]).

I_p , mA	T , $^\circ\text{C}$	U_p , mV	U_H , mV
30	19.1	1481	35
..	..	..	..

Table 1: An example of the table for zero field measurements

Part 2: Room temperature experiment

1. Turn off the AC/DC power supply.
2. Connect DC power output to the coils mounted on the core: connect the red lead from power supply **DC+** to the front top coil socket, then connect front lower coil socket with back lower coil socket, and close the circuit by connecting the back top coil socket to the **DC-** power output using the blue lead. This wiring produces a magnetic field B flow from front to back.
3. Turn on the Teslameter and zero it.
4. Connect the probe to the Teslameter, take off its cap, and accurately insert it into the module through the hole on top. Make sure it faces the front. BE CAREFUL not to harm neither the semiconductor sample, nor the probe.
5. Add the polar pieces carefully. The system mounted and wired is shown on the Fig.6.
6. Turn on the power supply.
7. Set the magnetic field to $B = 300$ mT by control knobs **V** and **A** on the power supply. Record readings for different sample currents I_p ranging from -30 mA to 30 mA in steps of about 5 mA. See a sample in the Table 2.
8. Set the current to $I_p = 30$ mA. Record readings for different magnetic field values B ranging from -300 mT to 300 mT in steps of about 50 mT. To obtain a negative magnetic field, swap the current leads on the power supply. Add records to the Table 2.
9. Apply misalignment correction (Eq. 18) to the Hall voltage.
10. Plot the Hall voltage U_H as a function of the sample current I_p and calculate the Hall coefficient:

$$R_H = \frac{U_H \cdot d}{B \cdot I}.$$

Determine the type of charge carriers in the semiconductor sample by R_H sign: negative for n-type, positive for p-type.

11. Calculate the density of the majority charge carriers n (the doping) using:

$$R_H = \frac{1}{nq}$$

and assuming that one type of carriers dominates (n here can be the density of either electrons or holes, depending on the sign of q). **Do calculations with units** to avoid mistakes.

$B,$ mT	$I_p,$ mA	$T,$ °C	$U_p,$ mV	$U_H,$ mV
300	30	19.1	1481	100
..	..	..	..	..

Table 2: An example of the table for room temperature measurements

12. Calculate the mobility of the majority charge carriers using:

$$\mu = \frac{|R_H|}{\rho} = \left| \frac{U_H}{U_p B} \right| \frac{W}{L}.$$

Part 3: Heating experiment

1. Set the current to $I_p = 30 \text{ mA}$ and the field to $B = 300 \text{ mT}$.
2. **Take the Hall probe OUT of the module.**
3. Start heating by pressing the **on/off** button on the backside of the module. The heating stops when the temperature reaches 140°C . Measurements must be taken during cooling, because temperature changes too rapidly during heating for the thermometer to be accurate.
4. Once heating stops, record temperature and voltage during cooling until the temperature reaches $40^\circ\text{C} - 50^\circ\text{C}$. Collect at least 10 points in range $140^\circ\text{C} - 120^\circ\text{C}$ and at least 15 points below 120°C . See the sample Table 3. Save the data.
5. Convert temperature to Kelvin. Plot $\log(U_p)$ as a function of $1/T$. Identify the linear regime (and therefore close to the intrinsic one) and determine the energy gap E_g using:

$$\sigma = \sigma_0 \cdot e^{-\frac{E_g}{2k_B T}}$$

6. Apply misalignment correction (18) and plot Hall voltage $U_H(T)$. For p-type samples the Hall voltage should cross zero, as expected by Eq.(15).

$T,$ $^\circ\text{C}$	$U_p,$ mV	$U_H,$ mV
..	..	..

Table 3: An example of the table for heating and cooling measurements.

References

- [Kasap(2006)] Safa O. Kasap. *Principles of Electronic Materials and Devices*. McGraw-Hill, New York, NY, 3rd edition, 2006. ISBN 978-0072957914.
- [Taylor(1996)] John R. Taylor. *An Introduction to Error Analysis: The Study of Uncertainties in Physical Measurements*. University Science Books, 1996. ISBN 093570275X.